# Electromagnetic Waves

⇒ EM Theory

# Introduction to gradient, divergence, curl :

→ Operator ($\nabla$) nabla)

$$\nabla = \frac{\partial \hat{i}}{dx} + \frac{\partial \hat{j}}{dy} + \frac{\partial \hat{k}}{\partial z}$$

It is useful in finding three qualities which arise in practical applications and are known as gradient, divergent, curl.

- Gradient : Let $\phi(x, y, z)$ be defined and differentiable at each point $(x, y, z)$ in certain region of space. ($\phi$ defines a differentiable scalar field)

$$\text{Gradient of } \phi(x,y,z) = \nabla \phi$$

$$= \left( \hat{i}\frac{\partial}{dx} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z} \right) \phi$$

$$= \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{dz}\hat{k}$$

- Divergence : Let $V(x, y, z) = V_1 \hat{i} + V_2 \hat{j} + V_3 \hat{k}$ be defined and differentiable at each point $(x, y, z)$ in certain region of space
($V$ defines a differentiable scalar field)

Divergence of $V(x,y,z) = \nabla V$

$$= \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right) \cdot (V_1\hat{i} + V_2\hat{j} + V_3\hat{k})$$

$$= \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}$$

$[\nabla \cdot V \neq V \cdot \nabla]$

- Curl : Let $V(x,y,z) = V_1\hat{i} + V_2\hat{j} + V_3\hat{k}$ be defined and differentiable at each point $(x,y,z)$ in certain region of space.
(V defines a differentiable vector field)

Curl of $V = \nabla \times V$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_1 & V_2 & V_3 \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial V_3}{\partial y} - \frac{\partial V_2}{\partial z}\right) - \hat{j}\left(\frac{\partial V_3}{\partial x} - \frac{\partial V_1}{\partial z}\right) + \hat{k}\left(\frac{\partial V_2}{\partial x} - \frac{\partial V_1}{\partial y}\right)$$

Q ① If $\phi(x,y,z) = 3x^2y - y^3z^2$
Find $\nabla\phi$ at point $(1,-2,-1)$
Ans: $-12\hat{i} - 9\hat{j} - 16\hat{k}$

② Find a unit normal to the surface $x^2y + 2xz = 4$ at point $(2,-2,3)$

③ $A = x^2z\,\hat{i} - 2y^3z^2\,\hat{j} + xy^2z\,\hat{k}$
$\nabla \cdot A = ?$ at point $(1,-1,1)$
Ans $= 3$

④ Find Curl $A = xz^3\,\hat{i} - 2x^2yz\,\hat{j} + 2yz^4\,\hat{k}$
$\nabla \times A = ?$ at point $(1,-1,1)$
Ans $= 3\hat{j} + 4\hat{k}$

→ Gauss Divergence Theorem

$$\oint_S (\vec{F}\cdot\hat{n})\,ds = \int_V (\nabla\cdot\vec{F})\,dV$$

It states that the surface integral of a normal component of a vector function over a closed surface is equal to the volume integral of the divergence of the vector function $\vec{F}$ over the volume 'V' enclosed by surface 's'.

→ Stokes Theorem

$$\oint_C \vec{A} \cdot \vec{dl} = \int_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{s}$$

It states that the line integral of vector field $\vec{A}$ around a closed group is equal to the surface integral of vector $\vec{A}$ taken over any surface S of which curve is a bounding edge.

# Laws of EM before Maxwell

1) Gauss's Law in electrostatics

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \quad \text{or} \quad \oint \vec{E} \cdot d\vec{s} = \frac{q}{\varepsilon_0}$$

2) Gauss's Law in Magnetostatics

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{or} \quad \oint \vec{B} \cdot \vec{ds} = 0$$

3) Faraday's law of induction

$$\vec{\nabla} \times \vec{E} = -\frac{\partial B}{\partial t} \quad \text{or} \quad \oint \vec{E} \cdot dl = -\frac{\partial \phi_B}{\partial t}$$

4) Ampere's Law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} \quad \text{or} \quad \oint \vec{B} \cdot \vec{dl} = \mu_0 I = \mu_0 \int_S \vec{J} \cdot dS$$

↳ Current density

- These 4 eqns are valid for steady state

# Displacement Current

According to Maxwell, it is not the current in a conductor that produces magnetic field but changing electric field in vacuum or dielectric also produces magnetic field. This means that changing electric field is equivalent to current and gives same effect to magnetic field as conductor current. This equivalent current is known as displacement current which can be expressed as $\varepsilon_0 \frac{d\phi_e}{dt}$

In case of parallel plate capacitor

$$i_c = \frac{dq}{dt} \rightarrow (1)$$

Here, electric disp in dielectrics is

$D = \varepsilon_0 E$

$$\boxed{\begin{array}{l} D = \sigma = q/A \quad \text{or} \\ q = \sigma A \ \text{ or } \ DA \end{array}} \rightarrow (2)$$

Putting (2) in (1)

$$i_c = \frac{d(D \cdot A)}{\partial t} = A\frac{\partial D}{\partial t} = A\frac{\partial(\varepsilon_0 \cdot E)}{\partial t}$$

$$= A\varepsilon_0 \frac{\partial E}{\partial t}$$

Maxwell suggested that term,

$$id = \varepsilon_0 \frac{d\phi_E}{dt}$$ ie Current inside dielectric

$$id = \varepsilon_0 \frac{\partial(AE)}{\partial t} = A\varepsilon_0 \frac{\partial E}{\partial t}$$

$$= A\varepsilon_0 \frac{\partial(D/\varepsilon_0)}{\partial t} = A\frac{\partial D}{\partial t}$$

$$\frac{id}{A} = \frac{dD}{\partial t} \Rightarrow Jd = \frac{\partial D}{\partial t} = \varepsilon_0 \frac{\partial E}{\partial t}$$

Hence, current arising due to time varying electric field b/w plates of capacitor is called displacement current